

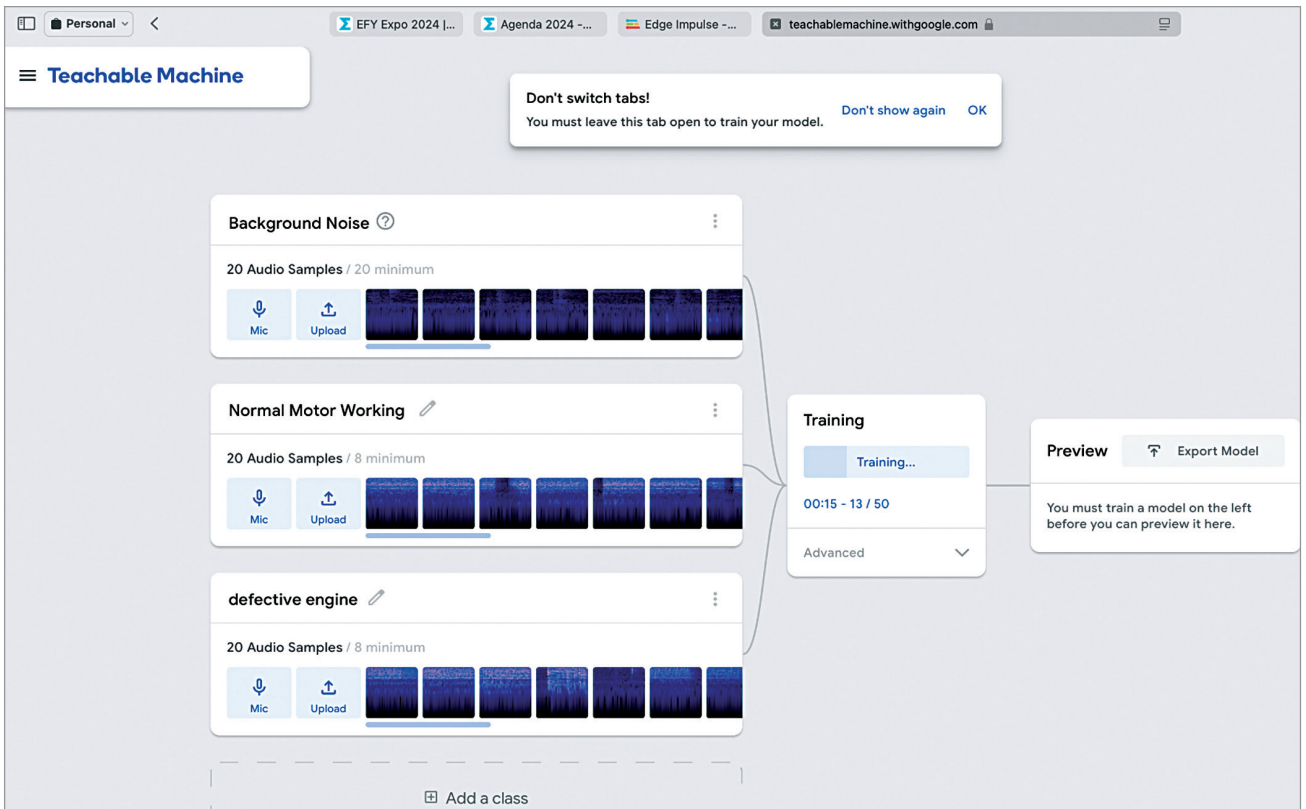


Fig. 3: Training the ML model in the Teachable Machine

```

    }

    async function init() {
      const recognizer = await createModel();
      const classLabels = recognizer.wordLabels();
      const labelContainer = document.getElementById("label-container");
      const alertDiv = document.getElementById("alert");

      for (let i = 0; i < classLabels.length; i++) {
        const label = document.createElement("div");
        label.innerText = `${classLabels[i]}: 0.00`;
        labelContainer.appendChild(label);
      }

      recognizer.listen(result => {
        const scores = result.scores;
        for (let i = 0; i < classLabels.length; i++) {
          const classPrediction = `${classLabels[i]}: ${scores[i].toFixed(2)}`;
          labelContainer.childNodes[i].innerHTML = classPrediction;

          // Check for "Defective Engine" threshold
          if (classLabels[i] === "defective engine" && scores[i] > 0.88) {
            alertDiv.style.display = "block"; // Show alert text
            fetch("http://192.168.230.48/alerton"); // Send alert to ESP32-S2
          }
        }
      });

      includeSpectrogram: true,
      probabilityThreshold: 0.75,
      invokeCallbackOnNoiseAndUnknown: true,
      overlapFactor: 0.50
    });
  }
</script>
</body>
</html>

```

Fig. 4: Snippet of the HTML code running the ML model and connection setup

Creating the AI model using Teachable Machine

1. Go to Teachable Machine
2. Click on 'Audio Project'
3. Record and label two categories:
 - 'Normal Engine' (record sounds of a healthy engine).
 - 'Defective Engine' (record sounds when faults occur like knocking and misfires.).
4. Train the model and export it as a TensorFlow.js model
5. Upload the ML model
6. Copy the model.json URL for use in our HTML file

Modifying the Teachable Machine code

The HTML file integrates the trained model and continuously monitors audio input. When an anomaly is detected, it sends an HTTP request to the IndusBoard Coin. The URL it